

HKCEE 1990
Mathematics II

Label 1 $(a^{2n})^3 =$

- A. a^{6n}
- B. a^{8n}
- C. a^{2n^3}
- D. a^{6n^3}
- E. a^{8n^3}

Label 2 $\frac{1 - \frac{x-y}{x+y}}{1 - \frac{x+y}{x-y}}$
Figure 2 Density 807

- A. $\frac{y-x}{x+y}$
- B. $\frac{x-y}{x+y}$
- C. $\frac{x}{y}$
- D. $\frac{x+y}{x-y}$
- E. $\frac{x-y}{x+y}$

Label 3 If $x = \frac{ab+1}{a-b}$, then $b =$

- A. $\frac{ax-1}{a+x}$
- B. $\frac{ax-1}{a-x}$
- C. $\frac{1-ax}{a+x}$
- D. $\frac{1-ax}{a-x}$
- E. $\frac{a+x}{ax+1}$

Label 4 If $f(n) = \frac{1}{2}n(n-1)$, then $f(n+1) - f(n) =$

- A. $f(1)$
- B. $f(n)$
- C. $\frac{n}{2}$
- D. 1
- E. n

Label 5 If $2 = 10^p$, $3 = 10^q$, express $\log \frac{1}{6}$ in terms of p and q .

- A. $-p-q$
- B. $\frac{1}{p}$
- C. $\frac{1}{p+q}$
- D. pq
- E. $p+q$

Figure 5 Density 1111

Label 6 Let $a \leq b > 0$. If a and b are respectively the 1st and 2nd terms of a geometric progression, the sum to infinity of the progression is

- A. $\frac{1}{a-b}$
- B. $\frac{a}{1-b}$
- C. $\frac{ab}{b-a}$
- D. $\frac{a^2}{a+b}$
- E. $\frac{a^2}{a-b}$

Label 7 $a^3 + 8a^{-3} =$

- A. $(a - \frac{2}{a})(a^2 + 2 + \frac{4}{a^2})$
- B. $(a - \frac{1}{2a})(a^2 + 1 + \frac{1}{4a^2})$

- C. $(a + \frac{1}{2a})(a^2 - \frac{1}{2} + \frac{1}{4a^2})$
 D. $(a + \frac{2}{a})(a^2 - 4 + \frac{4}{a^2})$
 E. $(a + \frac{2}{a})(a^2 - 2 + \frac{4}{a^2})$

Label 8 If p and q are the roots of the equation $x^2 - x + 3 = 0$, then $(2^{p-2})(2^{q-2}) =$

- A. $\frac{1}{32}$
 B. $\frac{1}{8}$
 C. $\frac{1}{2}$
 D. 8
 E. 32

Label 9 If $a : b = 3 : 4$ and $b : c = 2 : 5$, then $a^2 : c^2 =$

- A. $3 : 10$
 B. $9 : 25$
 C. $9 : 100$
 D. $36 : 25$
 E. $36 : 100$

Label 10 If 1 U.S. dollar is equivalent to 7.8 H.K. dollars and 1000 Japanese yen are equivalent to 53.3 H.K. dollars, how many Japanese yen are equivalent to 50 U.S. dollars?

- A. 1463
 B. 3417
 C. 7317
 D. 8315
 E. 20787

Label 11

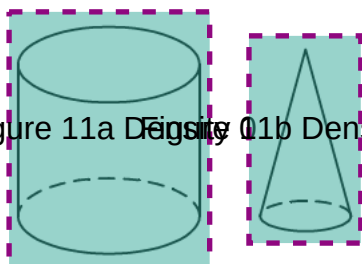


Figure 11a Density 201 Figure 11b Density 201

In the figure, the circular cylinder and the circular cone have the same height. The radius of the base of the cylinder is twice that of the cone. If the volume of the cone is 20 cm^3 , what is the volume of the cylinder?

- A. 40 cm^3
 B. 80 cm^3
 C. 120 cm^3
 D. 240 cm^3
 E. 300 cm^3

Label 12 The length, width and height of a cuboid are in the ratios $3 : 2 : 1$. If the total surface area of the cuboid is 88 cm^2 , find its volume.

- A. 6 cm^3
 B. 48 cm^3
 C. $48\sqrt{2} \text{ cm}^3$
 D. $96\sqrt{2} \text{ cm}^3$
 E. 384 cm^3

Label 13

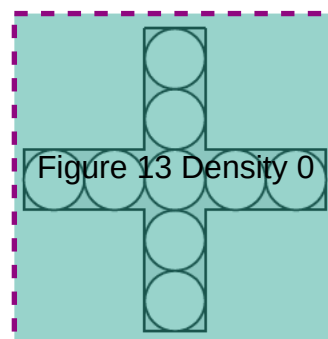


Figure 13 Density 0

In the figure, there are nine circles, each of radius 1. Find the shaded area.

- A. $9 - 9\pi$
 B. $36 - 9\pi$
 C. $40 - 9\pi$
 D. $10 - 10\pi$
 E. $40 - 10\pi$

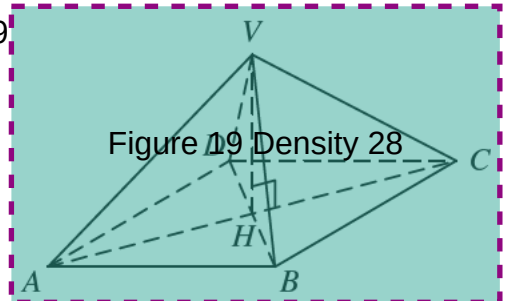
Label 14 Find the amount (correct to the nearest dollar) of \$10 000 at 12% p.a. compounded monthly, for 2 years.

- A. 10 201
- B. 12 400
- C. 12 544
- D. 12 697
- E. 151 786

- D. 2
- 5
- E. 2
- 5

Label 15 If a flat is sold for \$720 000, the gain is 20%. Find the percentage loss if the flat is sold for \$540 000.

- A. 5%
- B. $6\frac{1}{4}\%$
- C. 10%
- D. $11\frac{1}{9}\%$
- E. $33\frac{1}{3}\%$



The figure shows a right pyramid with a square base. VAB , VBC , VCD and VDA are equilateral triangles. Find $\sin \angle VAH$.

Label 16 $\sin(180^\circ + \theta) + \sin(\theta - 90^\circ) =$

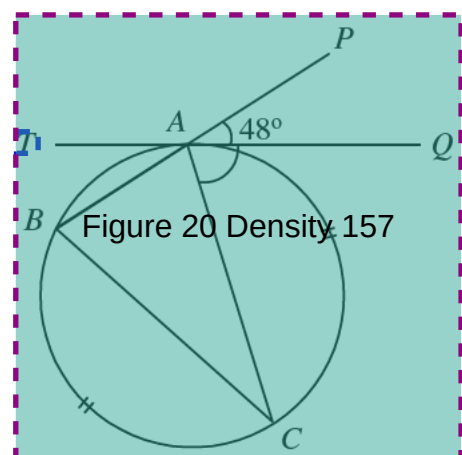
- A. $\sin \theta + \cos \theta$
- B. $\sin \theta - \cos \theta$
- C. $\cos \theta - \sin \theta$
- D. $-\cos \theta - \sin \theta$
- E. $2\sin \theta$

- A. 1
- 2
- B. 1
- 4
- C. 1
- $\sqrt{2}$
- D. 1
- $\sqrt{3}$
- E. $\sqrt{3}$
- 2

Label 17 If $0^\circ \leq x < 360^\circ$, which of the following equations has only one root?

- A. $\sin x = 0$
- B. $\sin x = \frac{1}{2}$
- C. $\sin x = 2$
- D. $\cos x = 0$
- E. $\cos x = -1$

Label 20



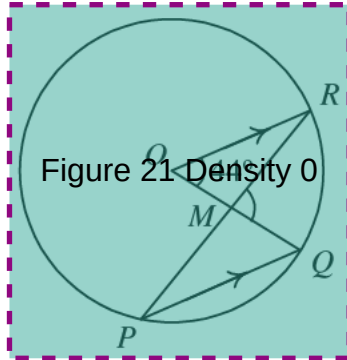
In the figure, TQ is the tangent to the circle at A . If arc $AC =$ arc BC and $\angle PAQ = 48^\circ$, then $\angle OAC =$

Label 18 If $\tan \theta = -\frac{4}{3}$ and θ lies in the second quadrant, then $\sin \theta - 2 \cos \theta =$

- A. 2
- B. -2
- C. 1
- 5

- A. 42°
- B. 48°
- C. 66°
- D. 71°
- E. 84°

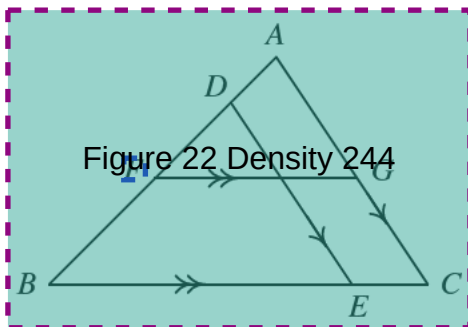
Label 21



In the figure, O is the centre of the circle. If $OR \parallel PQ$ and $\angle ROQ = 42^\circ$, find $\angle RMQ$.

- A. 21°
- B. 42°
- C. 63°
- D. 84°
- E. 126°

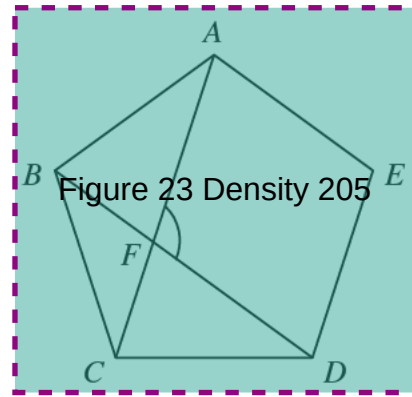
Label 22



In the figure, $AC \parallel DE$, $FG \parallel BC$ and $AD : DF : FB = 1 : 2 : 3$. If $BE = 10$, find FG .

- A. 5
- B. 6
- C. 8
- D. 9
- E. 10

Label 23



In the figure, $ABCDE$ is a regular pentagon. Find $\angle AFD$.

- A. 120°
- B. 112°
- C. 110°
- D. 108°
- E. 100°

Label 24 If the mean of the numbers 3, 3, 3, 3, 4, 4, 5, 5, 6, x is also x , which of the following is/are true?

- I. Mean = Median
- II. Mode = Range
- III. Median = Mode

- A. I and II only
- B. I and III only
- C. II and III only
- D. None of them
- E. All of them

Label 25 Ten years ago, the mean age of a band of 11 musicians was 30. One of them is now leaving the band at the age of 40. What is the present mean age of the remaining 10 musician?

- A. 40
- B. 39
- C. 37
- D. 30
- E. 29

Label 26
26. There are 7 bags, 3 of which are empty, and the remaining 4 each contains a ball. An additional ball is now put into one of the bags at random. After that a bag is randomly selected. Find the probability of selecting an empty bag.

- A. $\frac{2}{7}$
B. $\frac{3}{7}$
C. $\frac{6}{49}$
D. $\frac{12}{49}$
E. $\frac{18}{49}$

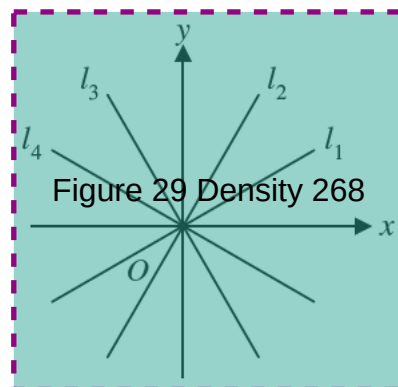
Label 27
27. $ABCD$ is a line segment. $AB : BC : CD = 3 : 2 : 1$. If $A = (4, 5)$, $D = (10, 11)$, find C .

- A. $(5, 6)$
B. $(6, 7)$
C. $(7, 8)$
D. $(8, 9)$
E. $(9, 10)$

Label 28
28. If the line $y = mx + b$ and $\frac{x}{a} + \frac{y}{b} = 1$ are perpendicular, find m .

- A. $\frac{a}{b}$
B. $\frac{b}{a}$
C. ab
D. $\frac{a}{b}$
E. $-\frac{b}{a}$

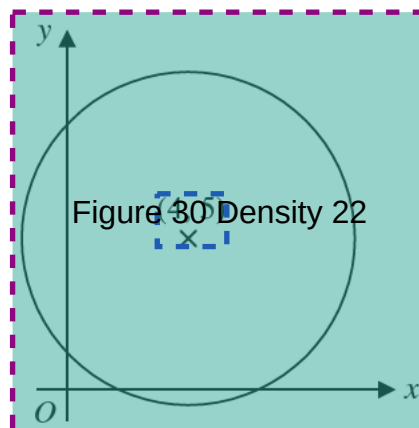
Label 29
29.



In the figure, the slopes of the straight lines l_1 , l_2 , l_3 , and l_4 are m_1 , m_2 , m_3 , m_4 respectively. Which of the following is true?

- A. $m_1 > m_2 > m_3 > m_4$
B. $m_2 > m_1 > m_3 > m_4$
C. $m_1 > m_2 > m_4 > m_3$
D. $m_2 > m_1 > m_4 > m_3$
E. $m_4 > m_3 > m_2 > m_1$

Label 30
30.



In the figure, a circle cuts the x -axis at two points 6 units apart. If the circle has centre $(4, 5)$, then its equation is

- A. $(x - 4)^2 + (y - 5)^2 = 25$
B. $(x - 4)^2 + (y - 5)^2 = 34$
C. $(x - 4)^2 + (y - 5)^2 = 52$
D. $(x + 4)^2 + (y + 5)^2 = 34$
E. $(x + 4)^2 + (y + 5)^2 = 25$

Label 31
31

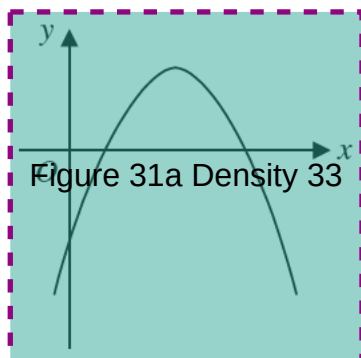


Figure 31a Density 33

The graph of $y = ax^2 + bx + c$ is given as shown. Which of the following is/are true?

- I. $a < 0$
- II. $b < 0$
- III. $c < 0$

- A. I only
- B. I and II only
- C. I and III only
- D. II and III only
- E. I, II and III only

90
32.

x	Sign of $f(x)$
1.22	+
1.23	+
1.24	+
1.25	-
1.245	+

Figure 31b Density 125

From the table, a root of the equation $f(x) = 0$ must be

- A. 1.20, correct to 2 decimal places
- B. 1.24, correct to 2 decimal places
- C. 1.25, correct to 2 decimal places
- D. 1.245, correct to 3 decimal places
- E. 1.2475, correct to 4 decimal places

Label 33
33

$$\frac{1}{1+\sqrt{2}} \times \frac{1}{\sqrt{2}+\sqrt{3}} \times \frac{1}{\sqrt{3}+\sqrt{4}} \times \frac{1}{\sqrt{4}+\sqrt{5}} =$$

- A. $\frac{1}{1-\sqrt{5}}$
- B. $\frac{1}{\sqrt{5}-1}$
- C. $1+\sqrt{5}$
- D. $1-\sqrt{5}$
- E. $-1+\sqrt{5}$

Label 34
34 Let $f(x) = 3x^2 - 4x + k$. If $f(x)$ is divisible by $x - k$, find the remainder when $f(x)$ is divided by $x + k$.

- A. $2k$
- B. k
- C. 0
- D. $-k$
- E. $-k-1$

Label 35
35 Let m be a constant. Find the value of x such that
$$\begin{cases} x^2 + x + 1 = m \\ x - 1 = \frac{26}{m} \end{cases}$$

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

Label 36
36 If $a < b < 0$, which of the following must be true?

- A. $a < -b$
- B. $\frac{a}{b} < 1$
- C. $a^2 < b^2$
- D. $10^a < 10^b$
- E. $a^{-1} < b^{-1}$

Label 37
37 The H.C.F. and L.C.M. of three expressions are xyz^2 and $x^3y^5z^4$ respectively. If two of the expressions are $x^2y^3z^3$ and x^3yz^2 , find the third expression.

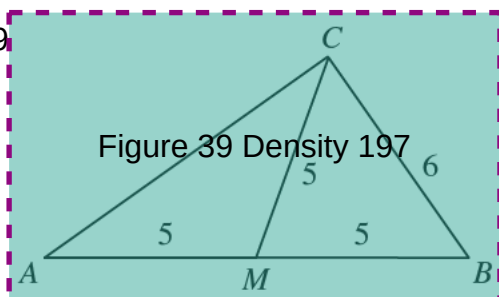
- A. $x^2y^3z^3$
- B. $x^2y^5z^3$

- C. xy^3z^3
 D. xy^5z^4
 E. xy^3z^4

Label 38
 38. Let a, x_1, x_2, b and a, y_1, y_2, y_3, b be two arithmetic progressions. $\frac{x_2 - x_1}{y_3 - y_2} =$

- A. $\frac{3}{4}$
 B. $\frac{3}{4}$
 C. $\frac{1}{4}$
 D. $\frac{4}{5}$
 E. $\frac{5}{4}$

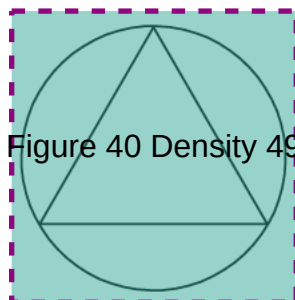
Label 39
 39.



In the figure, $AM = MB = MC = 5$ and $BC = 6$. The area of triangle $ABC =$

- A. 12
 B. 16
 C. 24
 D. 30
 E. 48

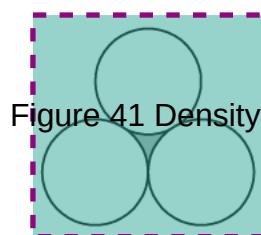
Label 40
 40.



In the figure, an equilateral triangle is inscribed in a circle of radius 1. The circumference of the circle is greater than the perimeter of the triangle by

- A. $4\pi - 3\sqrt{3}$
 B. $4\pi - \frac{3\sqrt{3}}{2}$
 C. $2\pi - \sqrt{3}$
 D. $2\pi - \frac{3\sqrt{3}}{2}$
 E. $2\pi - 3\sqrt{3}$

Label 41
 41.



Three equal circles of radius 1 touch each other as shown in the figure.

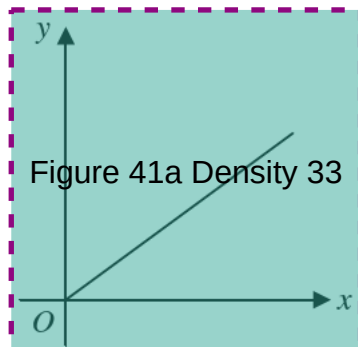
- A. $1 - \frac{\pi}{2}$
 B. $\sqrt{3} - \frac{\pi}{2}$
 C. $2\sqrt{3} - \frac{\pi}{2}$
 D. $\sqrt{3} - \frac{\pi}{6}$
 E. $2\sqrt{3} - \frac{\pi}{6}$

Label 42
 42. If A is 30% greater than B and B is 30% less than C , then

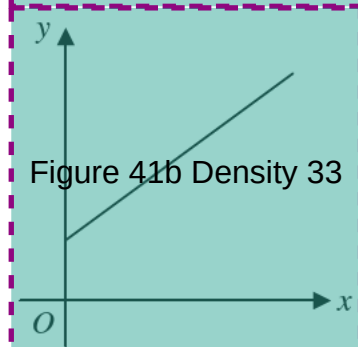
- A. A is 9% less than C
 B. C is 9% less than A
 C. $A = C$
 D. A is 9% greater than C
 E. C is 9% greater than A

Label 43
 43. Which of the following graphs shows that y is partly constant and partly varies inversely as x ?

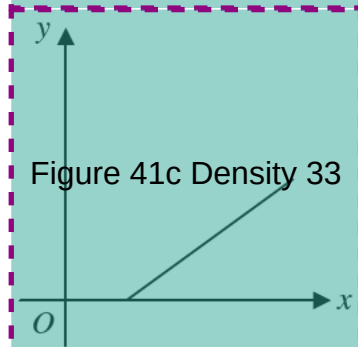
A



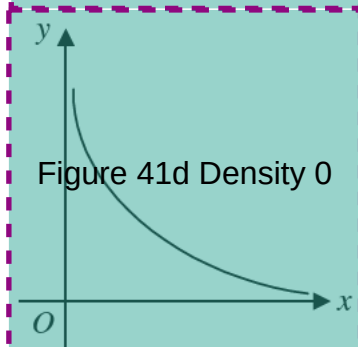
B



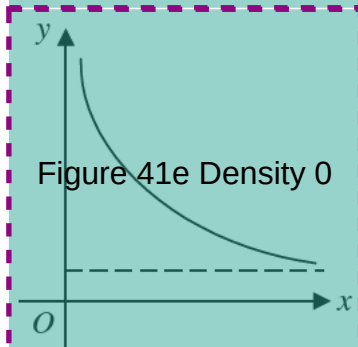
C



D



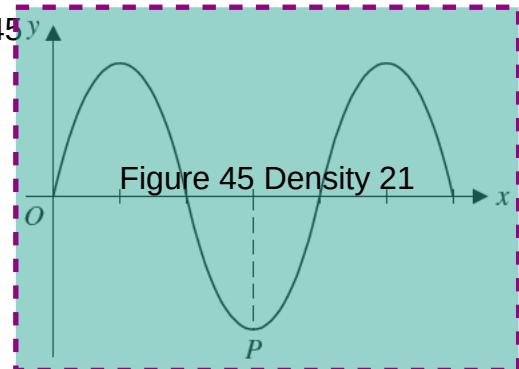
E



Label 44
44. If $\sin \theta$ and $\cos \theta$ are the roots of the equation $x^2 + k = 0$, then $k =$

- A. 1
- B. $\frac{1}{2}$
- C. $\frac{1}{4}$
- D. $\frac{1}{4}$
- E. $\frac{1}{2}$

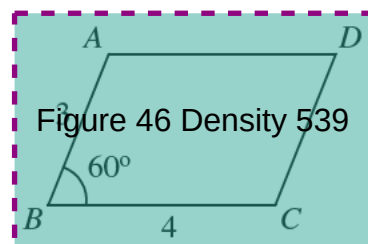
Label 45
45.



The figure shows the graph of $y = 3 \sin 2x$. The point P is

- A. $(\frac{4\pi}{3}, -3)$
- B. $(\frac{3\pi}{4}, -3)$
- C. $(\frac{4\pi}{3}, -1)$
- D. $(\frac{3\pi}{4}, -1)$
- E. $(\frac{3\pi}{2}, -1)$

Label 46
46.

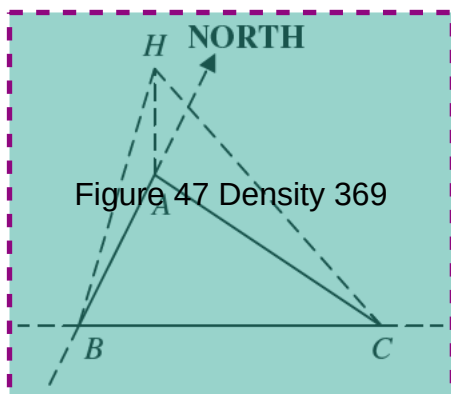


In the figure, $ABCD$ is a parallelogram. $BD =$

- A. 5
- B. 7
- C. $\sqrt{13}$
- D. $\sqrt{27}$
- E. $\sqrt{37}$

- A. 1
- B. $\frac{1}{2}$
- C. $\sqrt{2}$
- D. $\sqrt{3}$
- E. 2

Label 47

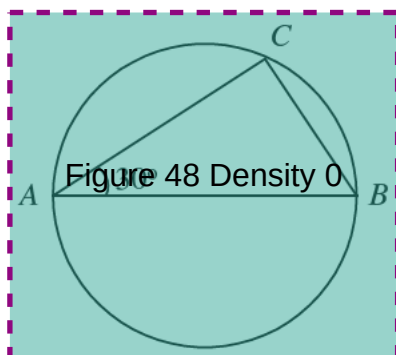


In the figure, A , B and C are three points on the same horizontal plane. A is due north of B , C is due east of B and H is a point vertically above A . Which of the following angles is/are 90° ?

- I. $\angle HAC$
- II. $\angle ABC$
- III. $\angle HBC$

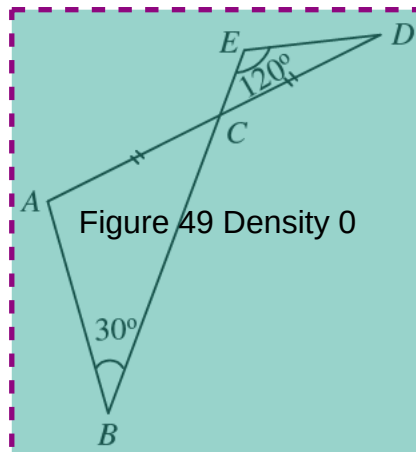
- A. I only
- B. II only
- C. I and II only
- D. I and III only
- E. I, II and III

Label 48



In the figure, AB is a diameter and $\angle BAC = 30^\circ$. If the area of $\triangle ABC$ is $\sqrt{3}$, then the radius of the circle is

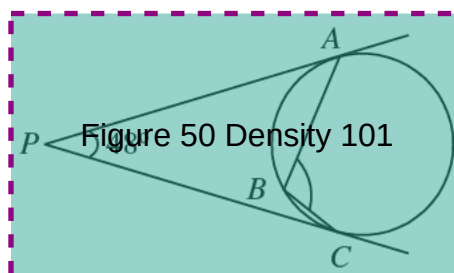
Label 49



In the figure, $AC = CD$, $\angle ABC = 30^\circ$ and $\angle CED = 120^\circ$. $\frac{AB}{DE} =$

- A. 1
- B. $\sqrt{2}$
- C. 1
- D. $\sqrt{3}$
- E. 2

Label 50



In the figure, PA and PC are tangents to the circle ABC . If $\angle P = 48^\circ$, then $\angle ABC =$

- A. 84°
- B. 96°
- C. 106°

- D. 114°
E. 132°

Label 51

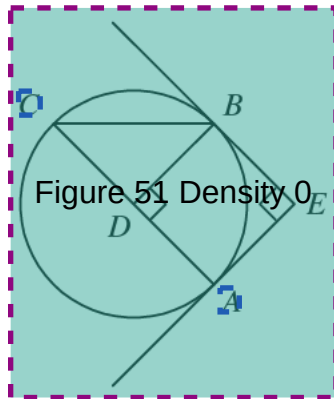


Figure 51 Density 0

In the figure, TA and TB are tangents to the circle ABC . If $TA \perp TB$ and $BD \perp AC$, find $\angle CBD$.

- A. 30°
B. 40°
C. 45°
D. 50°
E. 60°

Label 52

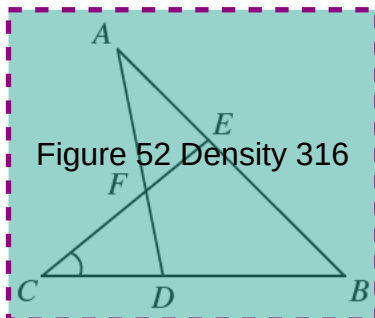


Figure 52 Density 316

In the figure, if $CD = CF$, $CE = BE$ and $DA = DB$, then $\angle C =$

- A. 30°
B. 36°
C. 40°
D. 45°
E. 60°

Label 53

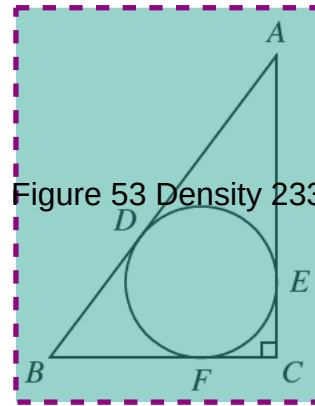


Figure 53 Density 233

In the figure AB , AC and BC are three tangents touching the circle at D , E and F respectively. If $AC = 24$, $BC = 18$ and $\angle ACB = 90^\circ$, find the radius of the circle.

- A. 3
B. 4
C. 5
D. 6
E. 7

Label 54

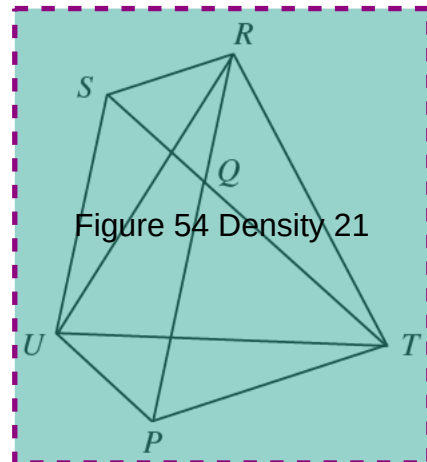


Figure 54 Density 21

In the figure, $\triangle PTQ$, $\triangle SQR$ and $\triangle RUT$ are equilateral triangles. Which of the following is/are true?

- I. $\triangle UPT \cong \triangle RQT$
II. $PU = QS$
III. $PQSU$ is a parallelogram

- A. All of them
B. None of them
C. I and II only
D. I and III only

E. II and III only